

Report 8: RGB–HSV Conversion

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1 Implementation

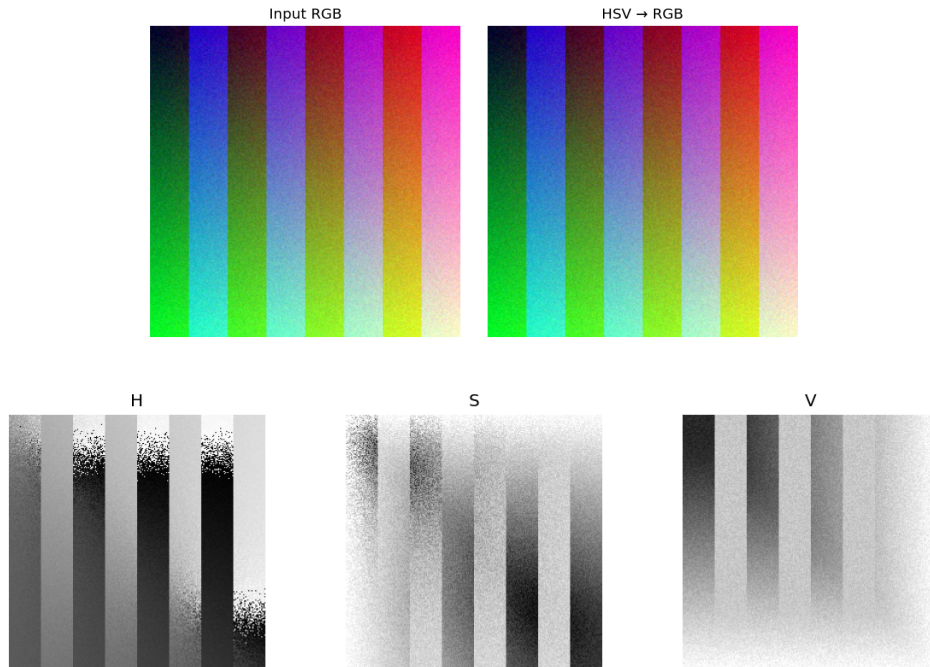
I converted interleaved RGB pixels into separate H, S and V arrays, then converted these arrays back to RGB. Each thread processes one pixel using a 16×16 block. H is stored in degrees; S and V are in $[0, 1]$. I compared HSV with Matplotlib and the reconstructed RGB with the input.

I ran the notebook on Google Colab with a Tesla T4 and a 256×256 sample image. GPU timings use `time.time()` after warm-up and synchronization, averaged over three runs; allocation and image transfers are excluded. Where measured, CPU time is one run; speedup is relative to that Python CPU implementation.

2 Results

Operation	GPU (ms)
RGB to HSV	0.1892
HSV to RGB	0.1217

Maximum normalized hue error: 7.09×10^{-8} . Maximum RGB reconstruction error: 0.



3 Discussion

The reconstructed RGB image exactly matched the input in this run. The separate H, S and V arrays allow later kernels to read one channel independently.